

products. If I buy a connector, then it should be able to go into ten different products. That is the standardisation done in the industry. In industry, use cases for mostly PCB-to-wire connectors can be found. Regarding other communications, especially CAN bus and RS 485, there is no specific standard for a connector. All can go in a screw terminal or 9-pin D-Sub terminal,” Mahendra Karandikar says.

From the perspective of PCB connectivity, there is no specific standardisation in the industry. BLC connectors, VGA connectors, and many others for PCB-to-wire connections are used.

The Arduino interface, which is popular for allowing modular design, can just be used for prototyping, and cannot be realistically extended to production requirements. “Arduino is kind of a processor not used in the industry. We use other versions of processors in the industry, which are embedded on the PCB. We use inter-PCB connectors for connection. We communicate using processors through SPI, UART, and I2C protocols,” says Abhinav Behl.

For high-speed data transfer at consumer end, the most popular is the USB. For industrial purposes, PCI and other high-speed protocols are used as well.

Limitations of USB

One of the limitations of USB is that it cannot be extended to longer lengths (cannot go beyond one or two metres). If an extension cable is used, then sometimes there could be a data loss. So, USB is typically fit for consumers. In industry, USB is used for short-term usage, like firmware upgrades, but certainly not for real-time data transfer.

Agreeing to this, Rajanikant Mohan says that “very often, the quality of the signal obtained through USB significantly deteriorates when the cable length increases. A lot of people don’t realise that and it could impact your design. In the PCB domain, you have the advantage of not using a cable; you can route it on the PCB. But externally, if you are using a cable, then it becomes complicated.”

Another thing noticed with Type-C is that they have become ‘smart cables.’ When you are connecting different devices, which can handle

different power delivery, there is the concept of electronically marked cables which have a Type-C controller embedded inside them that helps to negotiate this power. This smart cable is popular in high-end devices.

Some design issues

“My experience with mobile charger cables is that a lot of mobile phone users put a cover on their mobile. When you use a mobile cover, it adds 1-2mm extra length for the connector. Sometimes, if you plug in the connector, the connector doesn’t get fully inserted,” says Mahendra Karandikar. “Since mobile covers have been there in the market for several years, it is still unknown why USB OTG cable manufacturers aren’t making longer versions of the connector. So, even if I have 2-3mm thick mobile cover, it should get inserted completely. Many times, we have to remove the cover of the mobile phone and plug in the charger to it to ensure that it is inserted completely.”

“We faced a lot of trouble because the connector for our miniature wearable device took up a big chunk of its area (about 5mm × 4mm), due to which there was less space on the PCB for the charging circuitry, MCU, and antenna matching networks. We tried a lot of explorations by having magnetically attached cables and all sorts of paraphernalia. But we realised that there was no standard out there for any of this. In fact, to find a manufacturer who will make these miniature pluggable cables was almost impossible,” says Abhishek Satish.

He continues, “(In such a scenario) one is left with no option but to create one’s own connectors and cables. So, to reduce the total amount of space that had been taken, we opted for mid-mount USB. This was surface mount. So, it basically sat in between the PCB rather than on top or below it. That kind of saved us some space. But it came with its own set of challenges because now when you don’t have a through-hole element for a USB connector, the pad starts to fatigue (from all the constant plugging in and out). When you pull it out, it rips the USB connector from the pad, and you become stuck (as there is now no other way for charging). Probably we could have avoided it by increasing the pad

thickness that the PCB connectors sat on. But because the connector standard for the USB mini cable was little, there was no scope to expand the pads.

“To resolve the issue, the connector was mechanically supported from behind the PCB, so that if it was pushed in, it didn’t get pushed back and damage the pads. In the front, the enclosure was made very tight and there was no space for the connector to get accidentally pulled out.”

According to Abhishek, the physical design of the outer packaging also has an impact. “Most people don’t realise that the manufacturers give about 1-2mm gap to account for enclosure thickness. When the consumers used to plug in the cable, the connector it used to stick out; it felt like the connector had not gone in completely. So, they pushed it further because it looked to them it hadn’t gone in completely. We (too) didn’t account for that. And so, we had to end up increasing the enclosure thickness unnecessarily to make sure nobody forcibly pushed the connector and broke it in the process,” he says.

“In connectors, we have faced issues regarding PCB-to-wire connectors with a pitch distance of 2.5mm,” says Mahendra Karandikar. “When you want to pull out the connector while testing the PCB, or for replacement during design time, there is no other option but to hold the wires connected to it and pull it out. After inserting a few times, the wires get broken internally. In such connectors, there is a plug part which is soldered onto the PCB and the receptacle part which is inside the connector. So, when we insert the receptacle into the plug, a force is required to plug-in and plug-out the connector. There are two areas where force is required: the contact itself because the receptacle will have some surface area connection which gets connected to the other part. And there is the spring action for proper electrical contact. So that offers you some kind of resistance and requires force to pull out.”

He adds, “Another thing is that the plastic part of the connector has a locking mechanism, because once you insert a connector, it shouldn’t come out accidentally or due to vibration. So, a lot of force is required on account